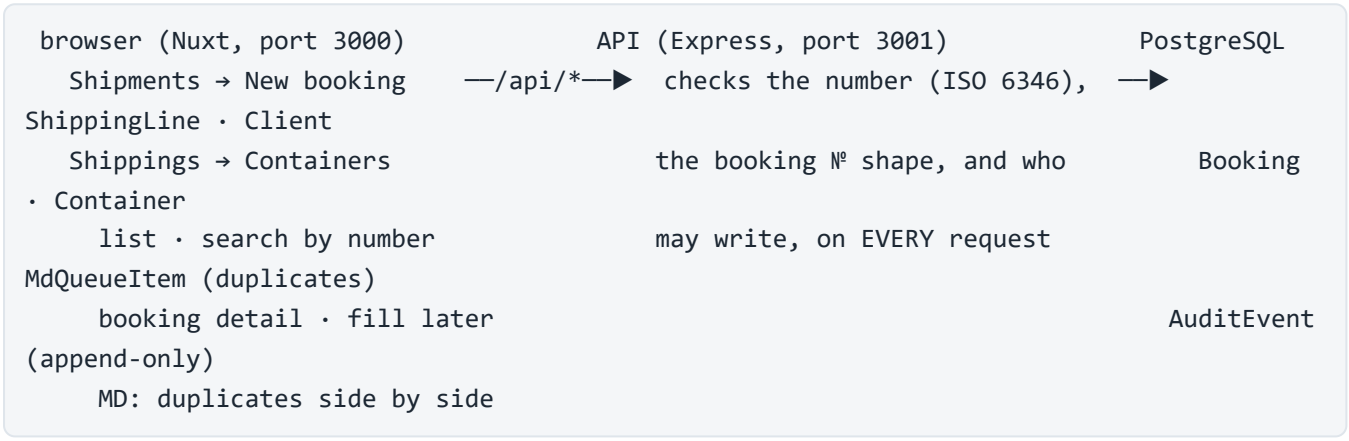
Slice 2 — Booking and container

Built: 23 October 2026 · **Status:** code complete, 22 automated tests passing (8 new), waiting for your live test. **Roadmap:** slice 2 of 12, the spine. A booking is a list of boxes and nothing more. No costs, no stages, no legs.

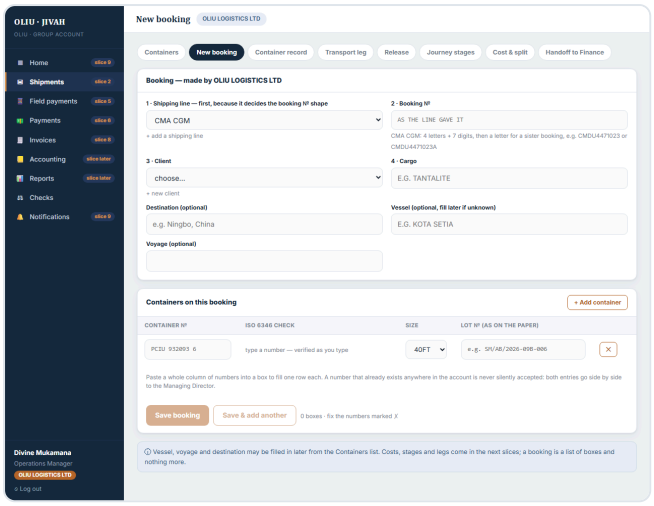
This note is written for you, not for a developer. It says what exists, how it works, how to try it, what was tested, what was left out, and the questions for you.

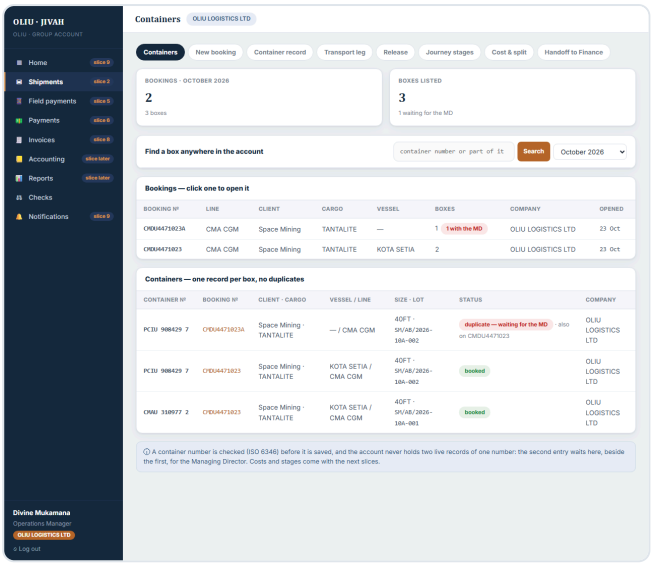
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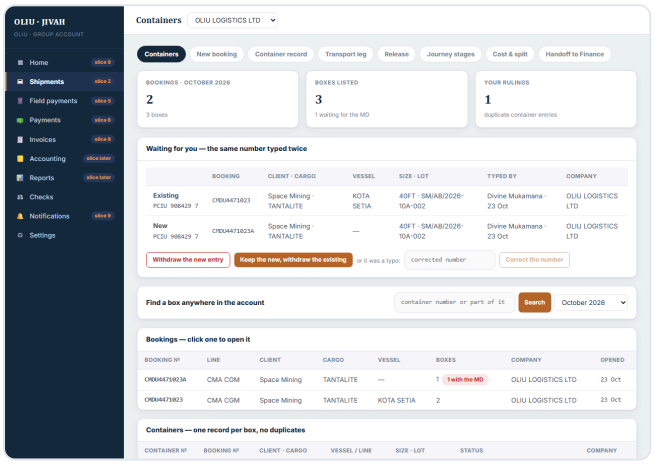
1. What exists now, in one picture

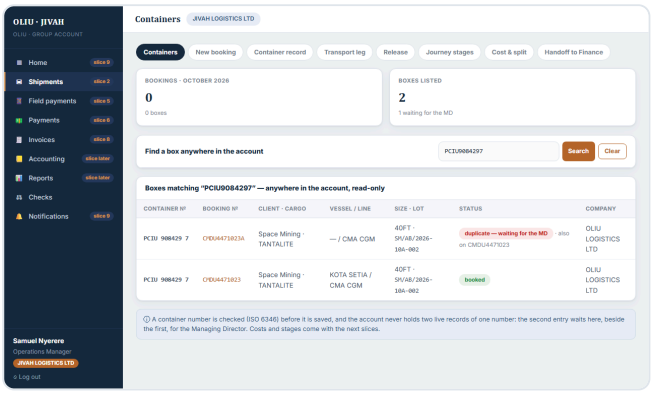


Two screens are new. The rest of the Shipments menu still says which slice brings it.


OLIU · JIVAH
OLIU · GROUP ACCOUNT
Home
Shipments
Field payments
Payments
Invoices
Accounting
Reports
Checks
Notifications
New booking
OLIU LOGISTICS LTD
Containers
New booking
Container record
Transport leg
Release
Journey stages
Cost & split
Handoff to Finance
Booking — made by OLIU LOGISTICS LTD
1 · Shipping line — first, because it decides the booking Nº shape
CMA CGM
+ add a shipping line
2 · Booking Nº
AS THE LINE GAVE IT
CMA CGM: 4 letters + 7 digits, then a letter for a sister booking, e.g. CMDU4471023 or CMDU4471023A
3 · Client
choose...
+ new client
4 · Cargo
E.G. TANTALITE
Destination (optional)
e.g. Ningbo, China
Vessel (optional, fill later if unknown)
E.G. KOTA SETIA
Voyage (optional)
Containers on this booking
+ Add container
CONTAINER Nº
ISO 6346 CHECK
SIZE
LOT Nº (AS ON THE PAPER)
PCIU 932003 6
type a number — verified as you type
40FT
e.g. SM/AB/2026-#06-#06
Paste a whole column of numbers into a box to fill one row each. A number that already exists anywhere in the account is never silently accepted: both entries go side by side to the Managing Director.
Save booking
Save & add another
0 boxes · fix the numbers marked X
Vessel, voyage and destination may be filled in later from the Containers list. Costs, stages and legs come in the next slices; a booking is a list of boxes and nothing more.
Divine Mukamana
Operations Manager
OLIU LOGISTICS LTD
Log out


OLIU · JIVAH
OLIU · GROUP ACCOUNT
Containers
OLIU LOGISTICS LTD
Containers
New booking
Container record
Transport leg
Release
Journey stages
Cost & split
Handoff to Finance
BOOKINGS · OCTOBER 2026
2
3 boxes
BOXES LISTED
3
1 waiting for the MD
Find a box anywhere in the account
container number or part of it
Search
October 2026
Bookings — click one to open it
BOOKING Nº
LINE
CLIENT
CARGO
VESSEL
BOXES
COMPANY
OPENED
CMDU4471023A
CMA CGM
Space Mining
TANTALITE
—
1
1 with the MD
OLIU LOGISTICS LTD
23 Oct
CMDU4471023
CMA CGM
Space Mining
TANTALITE
KOTA SETIA
2
OLIU LOGISTICS LTD
23 Oct
Containers — one record per box, no duplicates
CONTAINER Nº
BOOKING Nº
CLIENT · CARGO
VESSEL / LINE
SIZE · LOT
STATUS
COMPANY
PCIU 908429 7
CMDU4471023A
Space Mining · TANTALITE
— / CMA CGM
40FT · SM/AB/2026-10A-002
duplicate — waiting for the MD
also on CMDU4471023
OLIU LOGISTICS LTD
PCIU 908429 7
CMDU4471023
Space Mining · TANTALITE
KOTA SETIA / CMA CGM
40FT · SM/AB/2026-10A-002
booked
OLIU LOGISTICS LTD
CMAU 310977 2
CMDU4471023
Space Mining · TANTALITE
KOTA SETIA / CMA CGM
40FT · SM/AB/2026-10A-001
booked
OLIU LOGISTICS LTD
A container number is checked (ISO 6346) before it is saved, and the account never holds two live records of one number: the second entry waits here, beside the first, for the Managing Director. Costs and stages come with the next slices.
Divine Mukamana
Operations Manager
OLIU LOGISTICS LTD
Log out


OLIU · JIVAH
OLIU · GROUP ACCOUNT
Containers
OLIU LOGISTICS LTD
Containers
New booking
Container record
Transport leg
Release
Journey stages
Cost & split
Handoff to Finance
BOOKINGS · OCTOBER 2026
2
3 boxes
BOXES LISTED
3
1 waiting for the MD
YOUR RULINGS
1
duplicate container entries
Waiting for you — the same number typed twice
BOOKING
CLIENT · CARGO
VESSEL
SIZE · LOT
TYPED BY
COMPANY
Existing
PCIU 908429 7
CMDU4471023
Space Mining · TANTALITE
KOTA SETIA
40FT · SM/AB/2026-10A-002
Divine Mukamana · 23 Oct
OLIU LOGISTICS LTD
New
PCIU 908429 7
CMDU4471023A
Space Mining · TANTALITE
—
40FT · SM/AB/2026-10A-002
Divine Mukamana · 23 Oct
OLIU LOGISTICS LTD
Withdraw the new entry
Keep the new, withdraw the existing
or it was a typo:
corrected number
Correct the number
Find a box anywhere in the account
container number or part of it
Search
October 2026
Bookings — click one to open it
BOOKING Nº
LINE
CLIENT
CARGO
VESSEL
BOXES
COMPANY
OPENED
CMDU4471023A
CMA CGM
Space Mining
TANTALITE
—
1
1 with the MD
OLIU LOGISTICS LTD
23 Oct
CMDU4471023
CMA CGM
Space Mining
TANTALITE
KOTA SETIA
2
OLIU LOGISTICS LTD
23 Oct
Containers — one record per box, no duplicates
CONTAINER Nº
BOOKING Nº
CLIENT · CARGO
VESSEL / LINE
SIZE · LOT
STATUS
COMPANY


OLIU · JIVAH
OLIU · GROUP ACCOUNT
Containers
JIVAH LOGISTICS LTD
Containers
New booking
Container record
Transport leg
Release
Journey stages
Cost & split
Handoff to Finance
BOOKINGS · OCTOBER 2026
0
0 boxes
BOXES LISTED
2
1 waiting for the MD
Find a box anywhere in the account
PCIU9084297
Search
Clear
Boxes matching "PCIU9084297" — anywhere in the account, read-only
CONTAINER Nº
BOOKING Nº
CLIENT · CARGO
VESSEL / LINE
SIZE · LOT
STATUS
COMPANY
PCIU 908429 7
CMDU4471023A
Space Mining · TANTALITE
— / CMA CGM
40FT · SM/AB/2026-10A-002
duplicate — waiting for the MD
also on CMDU4471023
OLIU LOGISTICS LTD
PCIU 908429 7
CMDU4471023
Space Mining · TANTALITE
KOTA SETIA / CMA CGM
40FT · SM/AB/2026-10A-002
booked
OLIU LOGISTICS LTD
A container number is checked (ISO 6346) before it is saved, and the account never holds two live records of one number: the second entry waits here, beside the first, for the Managing Director. Costs and stages come with the next slices.
Samuel Nyerere
Operations Manager
JIVAH LOGISTICS LTD
Log out

2. The rules this slice makes true

RULE	WHERE IT LIVES	HOW YOU CAN SEE IT
A container number is checked before it is saved: four letters, six digits, one check digit (ISO 6346)	`shared/src/index.ts` → `checkContainerNumber`; the form uses it as you type, the API refuses a bad one	Type `PCIU9320935`: the field says "Check digit should be 6, not 5" and Save stays off
The account never holds two live records of one number	a database index that allows one ACTIVE record per number per account; a second entry is saved as DUPLICATE beside it	Enter a box that exists: both entries appear on the Containers screen, and in the MD's "Waiting for you" panel
Only the Managing Director rules on a duplicate	`POST /api/md-queue/:id/resolve` needs the `reconcile` permission, which only the MD holds	Divine sees the duplicate but no buttons; Eraste has three: withdraw the new, keep the new, correct the number
The person who makes the booking adds its boxes	writing needs the `book` screen and the `write` permission: MD and Operations Manager	Aline (Finance) has no Shipments menu; `/app/book` says the screen is not visible to her role
A booking belongs to the company that made it; the other side can read it, never change it	`mayTouchCompany` on every write; `mayEdit` in the booking detail	Samuel (JIVAH) finds an OLIU box by number, opens its booking, sees no edit form
The shipping line decides the booking № shape; its usual owner codes only produce a note	`ShippingLine.referencePattern` is enforced; `usualPrefixes` is advice	CMA CGM refuses `ABC123` and says "4 letters + 7 digits"; a `FFAU` box on PIL saves with a note
The browser never decides	menus, `mayEdit`, the duplicate ruling, the number check: all answered by the API	Change nothing in the browser, the API still decides

3. How a request travels through the code

Take "Divine, Operations Manager at OLIU, saves booking DAR600233400 with twelve boxes".

1. The browser has already said ✓ next to each number, using the same `checkContainerNumber` the API uses. It calls `POST /api/bookings` with the line, the booking №, the client, the cargo, the vessel and the twelve boxes.
2. `attachSession` builds the Scope: `{ accountId, userId, role: OPS_MANAGER, companyId: OLIU }`.
3. The route (api/src/modules/shipments/routes.ts) asks `requireScreen('book')` and `requirePermission('write')`. It checks every number again, checks the booking № against the line's shape, and that the № is new.

4. In one transaction: the booking row is written for company OLIU, then each box. For each box the route looks for a live record of that number in the account. None: the box is ACTIVE. One found: the box is saved as DUPLICATE, pointing at its twin, and an MD-queue item is opened. Every step is an audit row.
5. The answer lists the boxes, the duplicates (with the booking they already sit on) and the notes.

Take "Samuel, Operations Manager at JIVAH, looks up PCIU 932093 6".

1. `GET /api/containers?q=PCIU9320936`. Scope: `{ role: OPS_MANAGER, companyId: JIVAH }`.
2. Without `q` the route lists only JIVAH's bookings — none yet. With `q` it searches the whole account, because the box is the unit of truth and JIVAH will handle it after the hand-over.
3. The row comes back with its booking №, client, vessel, line and the company that made the booking. Opening the booking returns `mayEdit: false`; a PATCH or an added box from Samuel is refused with 403.

Take "Eraste rules on the duplicate".

1. `GET /api/md-queue` returns each open case with both entries: number, booking, client, cargo, size, lot, who typed it and when, and which company.
2. `POST /api/md-queue/:id/resolve` with `WITHDRAW_NEW`, `KEEP_NEW` or `CORRECT_NUMBER` + the number. One transaction changes the statuses, closes the case and writes `duplicate_resolved` to the trail.

4. Folders, so you can find things

```
shared/src/index.ts                 checkContainerNumber (ISO 6346), sizes, the booking
/ container / line /
                                    client / resolution inputs — one rule, read by the
API and the web app
api/src/modules/shipments/          routes.ts and README.md — everything of this slice
on the API side
api/src/middleware/auth.ts          requireScreen: a route belongs to a screen; only
roles that open it may call it
api/prisma/schema.prisma            ShippingLine, Client, Booking, Container,
MdQueueItem
api/prisma/migrations/20261023000000_slice2_booking_container/   the SQL, with the one-
live-record index at the end
api/prisma/seed.ts                  five shipping lines and the two clients of the
October invoices
api/test/slice2.test.ts             the roadmap's runs as code
web/components/ScreenBooking.vue    Shipments → New booking
web/components/ScreenContainers.vue Shipments → Containers: list, search, booking
detail, the MD's panel
```

5. Try it yourself

```
npm run db:start      (or the window titled "OLIU db:start" if it is still open)
npm run db:migrate
npm run db:seed       adds the shipping lines and the two clients; people untouched
npm run dev:api
npm run dev:web
```

Password for every demo person: `demo`. Codes land in http://localhost:3001/dev/mail.

SIGN IN AS	E-MAIL	ROLE IN THIS SLICE
Divine Mukamana	divine.m@oliulogistics.rw	Operations Manager, OLIU: makes the bookings
Eraste K.	eraste.k@oliulogistics.rw	MD: rules on duplicates; can also book, for the company in the switcher
Samuel Nyerere	samuel.n@jivahlogistics.co.tz	Operations Manager, JIVAH: the other side
Aline Uwase	aline.u@oliulogistics.rw	Finance Officer: no Shipments menu

6. Your test road — seven runs, tick each one

The two October invoices give the eighteen boxes. The invoices print the BL №, not the shipping line's booking №; use the BL № as the booking № for the test (question 1 below). Size is not on the Transvera invoice: use 40FT. Paste a column of numbers into the first container box and the form fills one row per number.

#	RUN	SIGN IN AS	WHAT MUST HAPPEN	✓
1	Space Mining	Divine	Shipments → New booking · line PIL · booking № `DAR600233400` · client Space Mining · cargo TANTALITE · vessel KOTA SETIA · the 12 boxes below with their lot numbers · every row shows ✓ · Save → "saved with 12 boxes"	☐
2	Transvera	Divine	booking № `DAR600210600` · client Transvera · the 6 boxes below · Save → 6 boxes · Containers list now shows 18 boxes and 2 bookings, this month	☐
3	A wrong digit	Divine	New booking, type `PCIU9320935` · the row says "Check digit should be 6, not 5" · Save stays off until you fix it	☐
4	A duplicate	Divine, then Eraste	New booking `DAR600241900`, box `FFAU8028326` (already on run 1) · it saves and says both entries went to the MD · as Eraste, Containers shows "Waiting for you" with both entries side by side · rule "Withdraw the new entry" · the row on DAR600241900 shows "withdrawn"	☐
5	The other side	Samuel	Containers: "No bookings this month on your side" · search `PCIU9320936` → found, with booking DAR600233400, Space Mining, KOTA SETIA, made by OLIU · click the booking № → read-only, no edit form	☐
6	Fill later	Divine	Containers → click DAR600233400 · voyage and destination · Save these details · add a box `PCIU9084297` → "Box added" · the audit trail shows booking updated and container added	☐
7	Not for Finance	Aline	no Shipments menu · typing `/app/book` in the address bar says the screen is not visible to your role	☐

Pass when eighteen real containers are in the system with no retyping, one wrong digit was caught, one duplicate reached the MD with both records shown, and nothing looked wrong, slow or confusing, wording included.

Space Mining, KOTA SETIA, 40FT — box and lot №

```
FFAU8028326   SM/AB/2026-09B-006        PIDU4544108   SM/AB/2026-09B-012
PCIU9320936   SM/AB/2026-09B-007        FFAU8024608   SM/AB/2026-09B-013
PCIU8878266   SM/AB/2026-09B-005        PCIU9040977   SM/AB/2026-09B-014
GLDU9067500   SM/AB/2026-09B-009        PCIU9154223   SM/AB/2026-09B-015
HPCU4743967   SM/AB/2026-09B-011        HPCU5173344   SM/AB/2026-09B-016
GAOU7500478   SM/AB/2026-09B-008
GCXU5105809   SM/AB/2026-09B-010
```

Transvera, KOTA SETIA, 40FT assumed — box and lot №

```
PCIU0016178   SFN/LIS/002/RH/SEP-26     PIDU2026877   SFN/LIS/003/RH/SEP-26
PCIU1809371   SFN/LIS/001/RH/SEP-26     PCIU2892210   SFN/LIS/006/RH/SEP-26
PCIU1619134   SFN/LIS/004/RH/SEP-26     GCXU5178913   SFN/LIS/005/RH/SEP-26
```

7. What was tested, and how

Twenty-two automated tests run against a fresh database on every `npm test` (about 15 seconds); eight are new:

- The eighteen real numbers of the two invoices pass the check; `pciu 932093-6` is read as `PCIU9320936`.
- A wrong check digit is refused with the digit it should be; a wrong shape and a wrong fourth letter are refused.
- The two invoices enter as two bookings with eighteen ACTIVE boxes, lot numbers kept; the bookings list counts them.
- The API refuses a wrong digit even if the browser did not, and saves nothing of that booking.
- A number that exists is saved as a duplicate beside its twin; exactly one live record per number remains; the Operations Manager cannot rule; the MD sees both sides and each of the three rulings does what it says; a ruling cannot be given twice; a correction to a number that is also taken is refused; the trail has every step.
- JIVAH's own list is empty, a search finds the OLIU box with its booking and client, read-only; the owner may fill the later fields and add boxes, and cannot add the same box twice.
- A line's booking № shape is enforced, its usual owner codes only produce a note, a line cannot be added twice under a different case, and an unusable pattern is refused.
- The MD books only for the company chosen in the switcher; a Finance Officer cannot book or add a client; another account sees none of it — not by search, not by id, not in the line list.

Also done by hand through the running web app: the two screens as Divine, Eraste and Samuel (the screenshots above), the as-you-type check, the paste of several numbers, the duplicate warning before saving, the three rulings.

Security checklist for this slice: every query scoped by account, and by company where the record is company-bound · every input validated by the shared schema, numbers normalised before any comparison · the booking № pattern is typed by the MD/Operations Manager and run only against a 40-character field · the one-live-record rule is enforced by the database itself, not only by the code, so two people typing the same new number at the same moment cannot both win · nothing is deleted, a withdrawn box stays in the record · the MD-queue ruling needs the `reconcile` permission · dependencies unchanged.

8. What was deliberately left out

- No costs, no stages, no legs, no board view. The box's stage is BOOKED and stays so until slice 4.
- No screen of their own yet for shipping lines and clients: both are added from the booking form. Settings → Configuration (slice 3) will list them properly.
- The MD queue has no menu of its own yet: the duplicates panel lives at the top of the Containers screen for the MD. Checks → MD queue arrives with slice 6 and takes these cases

over.

- No "move boxes to a sister booking" tool. A sister booking is typed as a new booking; the line patterns accept the letter suffix (CMDU4471023A).
- No booking-level draft/final documents, no per-container packet: the documents slice, later, by the person who makes the booking (your decision of 23 October).
- No Excel download.

9. Things that changed on the way

- **The mockup's sample numbers are not real container numbers.** TCLU 1234565 and MSKU 6011240 fail the ISO 6346 check; all eighteen on your invoices pass it. The mockup will be corrected the next time it is touched.
- **Owner codes belong to leasing companies, not to the line.** FFAU, GLDU, HPCU, GAOU and GCXU boxes travel on PIL vessels in your own invoices. So a line's "usual codes" only produce a note, never a refusal.
- **Uniqueness is per account, not across all tenants.** The same leased box can legitimately appear in another company's system in another month. "Unique system-wide" on the roadmap means within your account.
- **Sister bookings.** The CMA CGM, Maersk and Hapag-Lloyd patterns accept one letter after the number.
- **The screenshots** came from a demo booking CMDU4471023 and its sister CMDU4471023A with one repeated box, removed afterwards so your test road starts clean. The audit trail keeps those rows, as it must.

10. Questions for you

1. **The booking №.** The invoices print PIL's BL numbers (DAR600233400 …), not booking numbers. What does PIL's booking confirmation number look like? Then PIL gets a shape too, and the hint under the field says it.
2. **Who owns the lists** of shipping lines and clients: the Operations Manager, as now, or only you?
3. **JIVAH's working list** is empty until the hand-over flip of slice 4; they find boxes by number meanwhile. Keep it that way, or should JIVAH see every box of the account from the day it is booked?
4. **Container size** is missing from the Transvera invoice. Is 40FT right?
5. The three questions of slice 1 are still open: the department mailboxes, the Reporting Officer's scope, the real list of twelve people.